Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

What course should I study? Investigating the application of a Recommender System for Irish college courses

Brendan McCann

A Dissertation

Presented to the University of Dublin, Trinity College

in partial fulfilment of the requirements for the degree of

Masters in Computer Science

Supervisor: Owen Conlan

April 2026

Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

Brendan McCann

February 18, 2026

What course should I study? Investigating the application of a Recommender System for Irish college courses

Brendan McCann, Masters in Computer Science

University of Dublin, Trinity College, 2026

Supervisor: Owen Conlan

...ABSTRACT... **TODO**

Acknowledgments

TODO

Brendan McCann

University of Dublin, Trinity College
April 2026

Contents

List of Tables

List of Figures

Chapter 1

Introduction

1.1 Motivation

In 2024, 79% of Irish secondary school students progressed to college after graduation.[1] [2] The students who have considered progressing to college have all been posed with the question: *What course should I study?*

This is a daunting question for many reasons. The majority of secondary school students are faced with making this decision at 16-19 years old, an age with limited life experience and knowledge about the world. This decision comes with many personally confronting questions such as: *"What do I want to do with the rest of my life?"* and *"What do I love doing?"* among many others. There also exists the fear of choosing a college course the student regrets - in 2018, 31% of recent Irish college graduates reported they were not likely to study their college course again, [3] adding more weight to this important life decision.

Choosing a college course is not only a challenging decision to wrestle with as an individual. Students are shown to be influenced by a multitude of external factors when choosing their college course. For instance, parents are one of the most influential factors on a student's college course choice. [4] In addition to this, an individual's gender, socioeconomic status and their parent's educational background are among the most prominent societal influences on an individual's college course choice [4]. In short, choosing a college course is a difficult decision as the student must navigate their personal preferences and societal influences.

A Recommender System (RS) is a technology that recommend items to a user. Well-known examples of RS's include Netflix [5] recommending films to their customers, or Amazon [6] suggesting products to their customers. One of the earliest known RS's was *Tapestry* in 1992, a system that made tailored mailing list recommendations to its users. [7] Around this time, the field of RS's received considerable interest as e-commerce websites could increase their revenue if they suggested more personalised products to their users. Amazon and Netflix are the most notable adopters of RS's in order to increase company revenue. [8]

The two most common types of RS's are *Collaborative Filtering* and *Content-based Filtering* RS's. [9] *Collaborative Filtering* recommends items to users based on what other similar users also liked, whereas *Content-based Filtering* recommends items to users based on the user's preferences.

As previously stated, an RS recommends *items* to a *user*. However, we can re-conceptualize an RS in the provided context of college courses: *college courses* can take the place of *items*, and a *student* can take the place of a *user*. Given the established difficulty in choosing a suitable college course, and a technology (RS) that could be appropriately applied to this domain, an exciting opportunity presents itself: A Recommender System for Irish College Courses. A technology, such as an RS, could aid a student in navigating a difficult decision-making process. This ultimately describes the motivation behind this research.

1.2 Research Question

In light of the motivation behind this research, the Research Question for this dissertation **TODO** can be formulated: *How can a Recommender System assist an individual in their college course decision-making process?*

1.3 Research Objectives

With the motivation and research question formulated, the research objectives for this dissertation are outlined below:

1. A
2. B

1.4 Overview

Chapter 2

State of the Art

At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

From the perspective of this document forming part of your degree, this chapter should demonstrate to the reader your knowledge of the area of your dissertation project. It should present your knowledge in a coherent and detailed form. The reader should understand that you have in-depth knowledge of the area of the dissertation without being overloaded with information.

2.1 Background

A section on the background of the dissertation should provide the reader with an introduction to existing technologies and concepts that form the basis of the work presented in the dissertation.

2.2 Closely-Related Work

Work in research areas tends to address a number of specific aspects. Ideally, the discussion of published research should focus on the aspects that have been addressed by various publications - and not a discussion of the individual publications.

For example, if the topic would be a discussion of work on programming languages, the subsections of the related work could be discussions of object orientation and its realisation in various languages or the use of lambda functions by these languages.

2.2.1 Aspect #1

2.2.2 Aspect #2

2.3 Summary

Summarize the chapter and present a comparison of the projects that you reviewed.

	Aspect #1	**Aspect #2**
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 2.1: Caption that explains the table to the reader

Chapter 3

Design

At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

3.1 Problem Formulation

This section should provide the reader with an overall description of the problem that will be addressed in the dissertation. In contrast to a generic discussion of the dissertation topic in the Introduction chapter, this section should provide a detailed discussion of the problem that has been identified based on the existing work that has been discussed in the preceeding chapter.

In some dissertations, it may make sense to convert this section into a short chapter of its own which follows the discussion of the existing work and preceeds the discussion of the work of the dissertation.

3.1.1 Identified Challenges

This section should present a short description of the gaps in the existing work and the relationship of these gaps to the work described in this dissertation.

3.1.2 Proposed Work

This section should provide a thorough description of the problem and an overview of the work proposed to address the problem.

3.2 Overview of the Design

A description of the approach that addresses the problem identified above.

3.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 4

Implementation

Guess what? At the beginning of each chapter, a description should introduce the reader to the content of the chapter. The description should explain to the reader the layout of the chapter, the contribution that the chapter makes to the overall dissertation and the contribution of the individual sections towards the overall chapter.

4.1 Overview of the Solution

The code in listing **??** is a demonstration how to include a file with code into the template.

4.2 Component One

The code in listing 4.1 is a demonstration how to include code in the template.

```
x = 1
if x == 1:
    # indented four spaces
    print("x is 1.")
```

Listing 4.1: Second Lengthy caption

4.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.

Chapter 5

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

5.1 Experiments

In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 5.1.

Column 1	**Column 2**	**Column 3**
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 5.1: Caption that explains the table to the reader

5.2 Results

Figures that present results such as figure **??** need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.

5.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 6

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter 5.

6.1 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

Bibliography

[1] Higher Education Authority. Key facts and figures for ireland's publicly-funded higher education institutions. https://hea.ie/statistics/data-for-download-and-visualisations/key-facts-figures-report/, 2024.

[2] Department of education press release: Ministers congratulate almost 61,000 students receiving their leaving certificate examination results today. https://www.gov.ie/en/department-of-education/press-releases/ministers-congratulate-almost-61000-students-receiving-their-leaving-certificate 2024.

[3] Higher Education Authority. Graduate outcomes: Honours degree graduates. https://hea.ie/statistics/graduate-outcomes-data-and-reports/graduate-outcomes-2018/honours-degree-graduates/, 2018.

[4] Oya Aydın. University choice process: A literature review on models and factors affecting the process. *Yuksekogretim Dergisi*, 5, 03 2015.

[5] https://www.netflix.com/.

[6] https://www.amazon.com/.

[7] David Goldberg, David Nichols, Brian M Oki, and Douglas Terry. Using collaborative filtering to weave an information tapestry. *Communications of the ACM*, 35(12):61–70, 1992.

[8] J Ben Schafer, Joseph Konstan, and John Riedl. Recommender systems in e-commerce. In *Proceedings of the 1st ACM conference on Electronic commerce*, pages 158–166, 1999.

[9] Charu C Aggarwal et al. *Recommender systems*, volume 1. Springer, 2016.